Technical specification template: Power Cables

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A6.
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Document control

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1. Scope

This template defines the technical requirements for power cables for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 60502 series (power cables with extruded insulation)
- IEC 60287 series (cable current rating calculation)
- IEC 60364 series (installations), including current-carrying capacity
- Local fire performance requirements [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Installation method (basis of the rating)	Single-core cables in trefoil on ventilated cable tray in air	PRELIMINARY_ASSUMPTION
Conductor operating temperature	90.0 degC	UNVERIFIED_PLACEHOLDER. Of the chosen insulation
Voltage drop limit	3.0 %	PRELIMINARY_ASSUMPTION. Project criterion
Cable sizes and runs	See the cable sizing workbook: illustrative selections only	UNVERIFIED_PLACEHOLDER
Tabulated current rating per run	[From the cable standard or manufacturer for the installation method; never assumed]	Pending
Fault withstand	[Conductor area to survive the calculated fault for the clearing time]	NOT CALCULATED

5. Technical requirements

5.1 Each cable shall carry its design current after applying the grouping, ambient and installation derating factors, using a tabulated rating for the actual installation method.

5.2 The cable shall keep the voltage drop within the project limit at the design current.

5.3 The conductor shall withstand the calculated fault current for the protection clearing time (adiabatic or manufacturer data).

5.4 Insulation, sheath, armour, fire performance (flame propagation, smoke and halogen emission) and identification: [project to define].

5.5 Parallel runs shall be of equal length and layout with the current sharing verified; bending radius and pulling tensions per the manufacturer.

6. Vendor data schedule

The tenderer shall complete the schedule below with its offer: the value offered, its unit, and the name and revision of the document that states it. A value without a source document is not accepted. The parameters are those in the manufacturer-data register; supplied values are imported into the calculations through the data override mechanism, and a value stays NOT VERIFIED until a reviewer records an attestation against its source (name, organisation, date, evidence reference, scope).

Parameter id	Parameter	Datasheet required	Current status	Value offered	Unit	Document and revision
cable_data.lv_reactance_ohm_per_km	LV cable reactance	Cable data sheet or the standard's current-rating table for the installation	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]

		method, with grouping and ambient factors				
cable_data.mv_reactance_ohm_per_km	MV cable reactance	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
cable_data.conductor_operating_temperature	Conductor temperature for resistance	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.CAB-MV-A.tabulated_rating_a	CAB-MV-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.CAB-GEN-A.tabulated_rating_a	CAB-GEN-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.CAB-UPS-IN-A.tabulated_rating_a	CAB-UPS-IN-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]

		grouping and ambient factors				
distribution.CAB-UPS-OUT-A.tabulated_rating_a	CAB-UPS-OUT-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
distribution.CAB-MECH-A.tabulated_rating_a	CAB-MECH-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
short_circuit.assumed_clearing_time_s	Assumed fault clearing time for cable and busduct withstand	Protection coordination study result and protective device time-current curves	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]

Model, manufacturer, country of manufacture, and the type-test reports for the offered design: [vendor to complete].

7. Compliance schedule

The tenderer shall state, for every standard in section 2 and every technical requirement in section 5, whether the offer complies. 'Complies' needs an evidence reference (type-test report, certificate, calculation, drawing or data sheet with revision). A statement without evidence is recorded as not demonstrated. The requirement references below are the numbered items of section 5 of this template; no clause of any standard is quoted.

Standard named in section 2	Edition offered	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
IEC 60502 series (power cables with extruded insulation)	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
IEC 60287 series (cable current rating calculation)	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
IEC 60364 series	[vendor to	[vendor to	[vendor to	[vendor to

(installations), including current-carrying capacity	complete]	complete]	complete]	complete]
Local requirements [project to add]	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]

Requirement (section 5)	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
5.1	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.2	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.3	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.4	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.5	[vendor to complete]	[vendor to complete]	[vendor to complete]

8. Inspection and quality requirements

8.1 Review of type-test certificates and drum test records.

8.2 Inspection of drum condition, markings and length.

8.3 Routine tests on every unit as required by the applicable standard and this specification.

8.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

8.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

8.6 Calibration certificates for test instruments.

9. Factory acceptance test (FAT) requirements

9.1 Routine tests per the standard: conductor resistance, voltage test, partial discharge for MV cables.

9.2 Witnessing: [project to define].

9.3 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9.4 Acceptance criteria are taken from the applicable standard and this specification [project to define]; a test passed against a criterion not stated in advance is not accepted.

9.5 Equipment shall not be shipped until the FAT report is accepted or a concession is recorded and approved by the owner.

10. Site acceptance test (SAT) requirements

10.1 Visual check of routing, bending radius and glands; insulation resistance and the specified electrical test on every cable after installation; phase and sequence checks.

10.2 A SAT procedure shall be submitted for approval before the tests. It states the test, the method, the instrument, the acceptance criterion and the witness, with the criteria taken from the applicable standard and this specification [project to define].

10.3 SAT is carried out on the installed equipment with its final connections, after the installation inspection and before commissioning loads are applied.

10.4 Results are recorded on signed sheets. Any deviation, retest and concession is listed with its disposition. A SAT is not passed while a deviation is open.

10.5 SAT results are compared with the FAT results; a significant difference is investigated and recorded.

11. Commissioning requirements

11.1 Insulation resistance and continuity; phasing.

11.2 MV cable test per the specification [project to define: very low frequency or DC withstand and partial discharge].

11.3 Verification of installed layout against the rating basis (spacing, grouping, burial or tray).

11.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

11.5 Integrated systems testing of the whole power chain (utility loss, generator start and transfer, UPS on battery, loss of one power path, return to normal) is carried out under a project procedure [project to define]. This equipment's commissioning is not complete until it has taken part in that test.

12. Submittals and documentation

Document	When required	Status
Completed vendor data schedule (section 6) and compliance schedule (section 7)	With the tender	[vendor to complete]
Datasheets and general arrangement drawings	With the tender, final before manufacture	[vendor to complete]
Type-test reports and certificates for the offered design	With the tender	[vendor to complete]
Inspection and test plan (ITP)	Before manufacture	[vendor to complete]
FAT procedure and report	Before and after the FAT	[vendor to complete]
Installation instructions and SAT procedure	Before delivery	[vendor to complete]
Operation and maintenance manuals, as-built drawings and final data	Before commissioning; as-built at handover	[vendor to complete]
Recommended spare parts and special tools lists	With the tender and at handover	[vendor to complete]
Training plan and materials [project to define]	Before commissioning	[vendor to complete]
Warranty terms [project to define]	With the tender	[vendor to complete]

13. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.

- The vendor data schedule (section 6) and the compliance schedule (section 7) are blank until a tenderer completes them.
- FAT, SAT and commissioning acceptance criteria are [project to define].
- This template has not been reviewed by an independent engineer.